



Faculty of Computer Studies and Information Technology
Department of Information Technology
IT 303 / IT 303L – SYSTEM INTEGRATION AND ARCHITECTURE 1
LABORATORY ACTIVITY 1 — First Semester, AY 2026-2027 Activity 1: A
responding loop, with its integration map

Intended Learning Outcome

ILO 1: Build the smallest complete loop that responds to the world, and draw its integration map.

Time allotment: 3 hours — the second laboratory session for this outcome. The first session covers the taught work, so ILO 1's 6 hours in the syllabus is the two sessions together.

Objectives

- Build a circuit on a simulated breadboard using the power rails correctly.
- Write a program with one input, one decision, one output and one line of reporting.
- Demonstrate the system responding from end to end, in both directions.
- Draw the integration map of what you built, naming every part and every junction.

Materials

- A browser and a free simulator account.
- Your device code from the parameter slip.
- One sheet of paper for the integration map.
- Tool for this activity: Tinkercad Circuits — free, browser, nothing installed.

Procedure

1. Create a new circuit and name it with your device code.
2. Place the controller board. Run a wire from its 5 V pin to the red power rail and from its GND pin to the blue rail.
3. Place an LED. Put a 220 ohm resistor in series with its longer leg. Wire the resistor to pin 13 and the shorter leg to the blue rail.
4. Place a push button across the central channel. Wire one side to the red rail. Wire the other side to pin 2, and from that same row put a 10k resistor down to the blue rail. That resistor is the pull-down, and without it the input reads unpredictably.
5. Open the code panel and switch it to text. In setup, set pin 2 as an input and pin 13 as an output, drive pin 13 LOW, and begin serial at 9600.
6. In loop, read pin 2 into a variable, decide whether it is HIGH, write that decision to pin 13, and print one line giving both the reading and the decision. Add a short delay.
7. Start the simulation. Press the button and hold it: the LED lights and the serial monitor prints a line. Release it: the LED goes out and another line prints.
8. Record both directions in your notebook — what you did, and what happened.

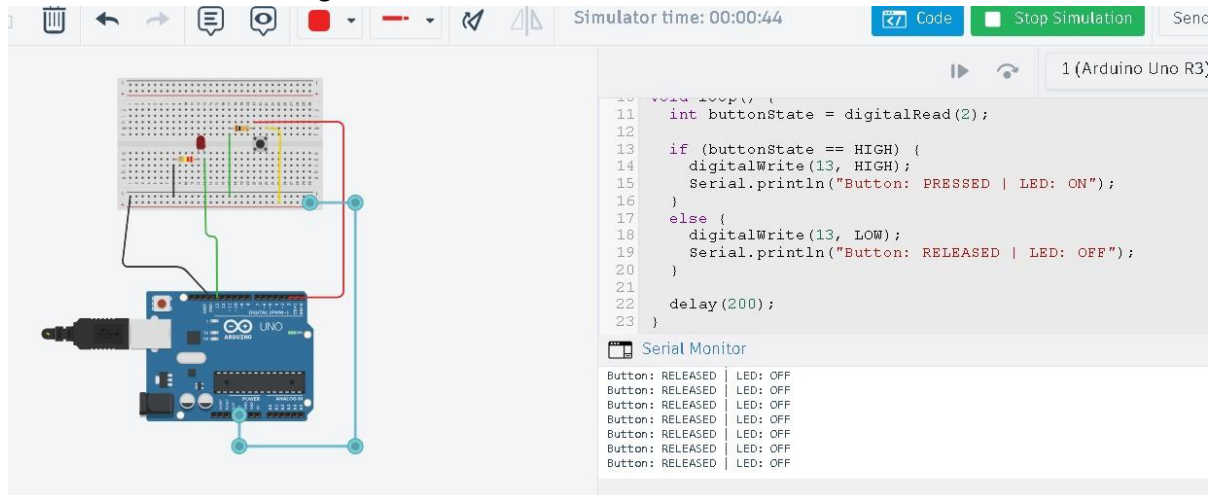




9. Draw the integration map: a box for each part, a line for each junction, and a label on each line saying what crosses it. You should have four boxes and three lines.

What to Submit

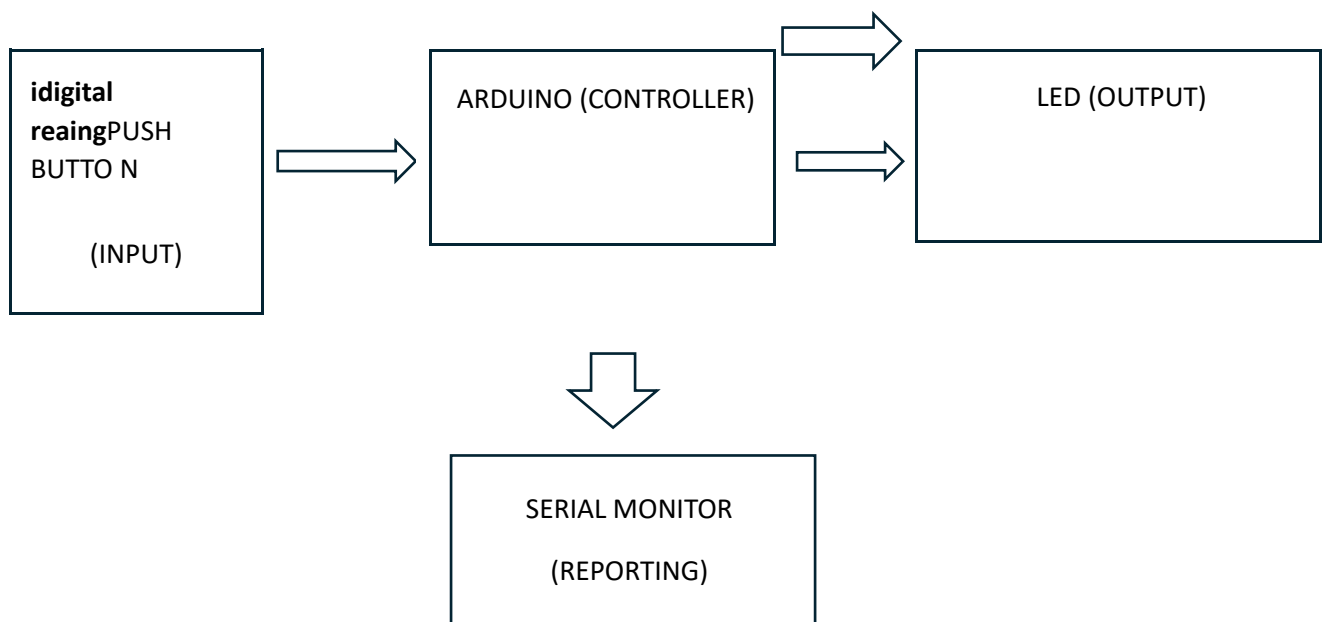
- A screenshot of the working circuit with the serial monitor visible.



- The two recorded lines of output, pressed and released.

```
Button: PRESSED | LED: ON
Button: PRESSED | LED: ON
Button: PRESSED | LED: ON
Button: RELEASED | LED: OFF
Button: RELEASED | LED: OFF
Button: RELEASED | LED: OFF
```

- Your integration map, drawn by hand or otherwise, with every line labelled.



How This Activity Is Marked

Rubric For Laboratory Activities, from the course syllabus. Each criterion is rated Excellent (4), Very

Satisfactory (3), Satisfactory (2) or Needs Improvement (1), and that share of its weight is awarded.

Criterion	Weight	What earns full marks here
Working System	30%	The system responds from end to end at the close of the session, and the part added this session works within it.
Program Quality	25%	The program is correct, readable and uses functions and variables appropriately.
Interface Note and Evidence	20%	Every junction made this session has an interface note giving signal, units, range, timing and absence behaviour, and the recorded readings support it.
Explanation	15%	Explains what each part does and why, including any fault encountered.
Timeliness and Compliance with Instructions	10%	Submitted within the session and follows every instruction given.

Reminders

- Everything runs in a browser on the laboratory computers. You buy nothing, and you install nothing.
- Save your own copy of every file. Work that exists only on a laboratory computer is not backed up.
- Evidence means a screenshot showing the thing AND the state you are claiming — the circuit running, the value readable, the serial line visible.
- Every session ends the same way: the system responds from end to end, and you have demonstrated it. A session that ends with a system that does not run is not finished, whatever else was built.
- Every junction you make gets an interface note, written at the bench while you make it. A note written later from memory records what you believe, not what is true.
- Your device code identifies your work. It is an opaque code and it is the only identifier that belongs in a project — never your name or your student number.

